Software Requirements Specification (SRS) for Vasudhar Mobile Application

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define and describe the functionality, features, constraints, and performance criteria of the Vasudhar Mobile App. This mobile application is designed for users in India to track their sustainability habits, receive personalized reports based on their behaviors, and engage with an AI-driven system that suggests improvements and encourages habit-forming practices through challenges, communities, and incentives.

1.2 Scope

This application will allow users to:

1. Sign up using Google or a traditional email/password mechanism.
2. Complete a sustainability questionnaire.
3. Receive a personalized sustainability report based on their responses.
4. Engage with a Language Learning Model (LLM) for explanations, suggestions, or general inquiries.
5. Participate in challenges related to sustainability and earn rewards for completing them.
6. Track their progress over time and refine estimates for continuous improvement.
7. Join communities to share experiences and tips with like-minded users.

1.3 Definitions, Acronyms, and Abbreviations

- **Sustainability**: Practices that reduce an individual's environmental impact.
- **LLM**: Large Language Model used for chatting, answering user queries, and suggesting improvements.
- **User**: A registered individual using the mobile app to track their sustainability habits.

- **Challenge**: Activities that users can participate in to improve sustainability habits.
- **Incentive**: Rewards provided for completing challenges.
- **Community**: Groups of users who share common interests related to sustainability.

1.4 References

- ISO/IEC 25010:2011 (Systems and Software Quality Requirements and Evaluation).
- IEEE Std 830-1998 (Recommended Practice for Software Requirements Specifications).

1.5 Overview

The SRS will cover the system's detailed requirements, both functional and non-functional. It will also specify the architecture, behavior, security, and usability aspects of the application.

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2. Overall Description

2.1 Product Perspective

The Vasudhar Mobile Application is a standalone application targeting Indian users. It will utilize AI-driven data analysis and recommendation engines, as well as LLM for interaction, and gamification techniques to motivate users to improve their sustainability habits.

2.1.1 System Interfaces

- **Database**: The system will use a cloud-hosted NoSQL database to store user data, responses, reports, and other relevant information.
- **AI Engine**: The backend analysis and recommendation system will use AI-based models to analyze user responses.
- **Google OAuth API**: For enabling Google sign-in functionality.
- **LLM API**: For querying the language model for real-time interactions with users.

2.1.2 User Interfaces

- **Mobile Interface**: A user-friendly mobile app (Android/iOS) with intuitive navigation to complete questionnaires, view reports, interact with LLM, participate in challenges, track progress, and join communities.

2.1.3 Hardware Interfaces

- Standard mobile devices (smartphones) with basic processing power and memory.

2.1.4 Software Interfaces

- **React Native Framework**: For mobile app development.
- **Backend Services**: A combination of Node.js, Python, and cloud services to support AI, LLM, and database functions.

2.1.5 Communications Interfaces

- **Internet Connection**: The mobile app will require constant internet access to send/receive data and interact with the backend and LLM services.

2.2 Product Functions

The app offers the following core functionalities:

1. **User Registration & Authentication**:
    - User can register via email or Google OAuth.
2. **Questionnaire Completion**:
    - Users are presented with a questionnaire regarding their sustainability habits (e.g., fuel usage, waste disposal).
3. **Sustainability Report Generation**:
    - An AI engine analyzes questionnaire responses to generate a comprehensive report on the user's sustainability performance.
4. **Personalized Recommendations**:
    - Tailored suggestions based on the user's sustainability report to encourage better habits.
5. **LLM Chat Interaction**:

    - Users can engage with an LLM to ask for explanations regarding their report, seek general advice, or discuss other sustainability-related topics.

6. **Progress Tracking & Refining Estimates**:
    - Users can periodically refine their estimates and re-answer questions to track progress over time.
7. **Challenges and Rewards**:
    - Users can participate in challenges (e.g., reducing water consumption) and earn incentives for successful completion.
8. **Community Interaction**:
    - Users can join communities and interact with other like-minded individuals for shared growth and support.

2.3 User Characteristics

- **Primary User**: Environmentally conscious individuals who want to track and improve their sustainability efforts.
- **Age Range**: 18-45 years.
- **Geographic Target**: Indian users, with options to expand globally.
- **Technical Expertise**: Basic smartphone users with internet access.

2.4 Constraints

- The app must comply with Indian data protection laws (e.g., Personal Data Protection Bill).
- Low bandwidth considerations for rural areas.
- Should be optimized for both iOS and Android platforms.

2.5 Assumptions and Dependencies

- Users will provide accurate information during the questionnaire for proper report generation.
- A consistent internet connection will be required to access all functionalities.

- The app assumes regular user engagement to generate meaningful reports and recommendations.

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3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Registration & Login

- **FR1.1**: The system shall allow users to sign up with an email and password or via Google OAuth.
- **FR1.2**: The system shall validate user credentials upon login.
- **FR1.3**: The system shall handle forgotten passwords through email recovery.

3.1.2 Sustainability Questionnaire

- **FR2.1**: The system shall present users with a dynamic sustainability questionnaire.
- **FR2.2**: The questionnaire shall include radio buttons, checkboxes, text, and numeric inputs.
- **FR2.3**: The system shall allow users to save their answers and submit once completed.

3.1.3 Report Generation & Analysis

- **FR3.1**: The system shall analyze the user's answers to generate a detailed sustainability report.
- **FR3.2**: The report shall include performance metrics, comparisons to average users, and specific areas for improvement.
- **FR3.3**: The system shall allow users to download or share their report.

3.1.4 Personalized Recommendations

- **FR4.1**: The system shall generate personalized sustainability recommendations based on the report.
- **FR4.2**: Recommendations shall be categorized (e.g., energy, waste, transport) and prioritized.

- **FR4.3**: The system shall notify users of new or improved recommendations based on updated inputs.

3.1.5 LLM Chat Interface

- **FR5.1**: The system shall allow users to chat with the LLM for report clarifications or general sustainability advice.
- **FR5.2**: The LLM should respond contextually based on user inputs and previous interactions.

3.1.6 Progress Tracking & Refining Estimates

- **FR6.1**: The system shall allow users to periodically refine estimates to track their sustainability progress.
- **FR6.2**: Based on refined estimates, the system shall generate updated reports and recommendations.

3.1.7 Challenges and Rewards

- **FR7.1**: The system shall allow users to participate in sustainability challenges (e.g., reducing plastic use).
- **FR7.2**: The system shall track the completion of challenges and offer rewards accordingly.

3.1.8 Community Interaction

- **FR8.1**: The system shall allow users to join sustainability-focused communities.
- **FR8.2**: The system shall allow users to post messages, share tips, and discuss their sustainability efforts.

3.2 Non-Functional Requirements

3.2.1 Usability

- **NFR1**: The app shall have an intuitive user interface that is easy to navigate.
- **NFR2**: The app shall support dark and light modes, auto-switching based on system settings.

3.2.2 Performance

- **NFR3**: The system shall handle up to 100,000 concurrent users.

- **NFR4**: The app shall load the sustainability report within 3 seconds on a stable connection.

3.2.3 Security

- **NFR5**: The system shall use SSL/TLS encryption for all data exchanges.
- **NFR6**: The app shall store sensitive data (e.g., user credentials) securely using encryption.

3.2.4 Availability

- **NFR7**: The system shall maintain 99.9% uptime, except for scheduled maintenance.

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4. System Architecture

4.1 Overview

The system architecture will follow a client-server model, with the mobile app acting as the client, and the backend (housed on cloud infrastructure) processing all analysis, data storage, and interactions with external APIs (e.g., LLM, Google OAuth).

4.2 Components

- **Client (Mobile App)**: Built using React Native to support both Android and iOS platforms.
- **Backend (Cloud-based API)**: Developed using Node.js for API endpoints and Python for AI/ML models for data analysis.
- **Database**: NoSQL cloud database for flexible storage of user data, reports, and preferences.

4.3 Security Architecture

- **OAuth 2.0**: Secure authentication for Google login.
- **Data Encryption**: Sensitive data will be encrypted at rest and in transit using AES-256 and SSL.

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5. Acceptance Criteria

1. The app must allow new users to sign up and receive their personalized sustainability report.
2. Users must be able to refine their estimates and view progress over time.
3. The LLM must provide relevant responses to user inquiries.
4. Users must be able to successfully complete challenges and receive incentives.
5. The system must function across different devices and operating systems, with good performance and usability.

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6. Appendices

6.1 Assumptions and Dependencies

- Third-party services such as Google OAuth and LLM API will remain functional and available.